

Exercise sheet 1-1.

February 24th

Due March 10th in class, to be combined¹ with sheet 1-2

Exercise 1. Let H be a p -adic Lie group. Let $\mathcal{C} = \mathbb{C}$ or $\mathcal{C} = \overline{\mathbb{Q}_\ell}$ with its usual topology. Let n be an integer and $\rho : H \rightarrow \mathrm{GL}_n(\mathcal{C})$ be a continuous group homomorphism. Show that $\ker(\rho)$ contains an open subgroup.

Exercise 2. Show that M_G is indeed a free module of rank 1 over $\mathbb{Z}[1/p]$. Show that any open compact subset $X \subset G$ can be written as a finite union of J -cosets $X = \bigsqcup_{i=1}^m g_i J$ for some $J \in \Omega(G)$. Show that the quantity $\nu(X) := m \cdot \nu(J) \in \mathbb{Z}[1/p]$ depends only on X and ν but not on the choice of g_i and J .

Exercise 3. Let $G = \mathrm{GL}_2(\mathbb{Q}_p)$, which acts on $\mathbb{P}^1(\mathbb{Q}_p)$ in a natural way. Consider the space $\pi = \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ on which G also acts. Show that π is generated by a single vector, i.e. there exists $v \in \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ such that

$$\mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p)) = \mathrm{span}_{\mathcal{C}} \langle g.v \mid g \in \mathrm{GL}_2(\mathbb{Q}_p) \rangle.$$

Remark 1. In the last exercise that, π has at most length 2 when $\mathrm{char}(\mathcal{C}) = 0$, and at most length 3 in general.

Our proof for Theorem A, B and C will actually work with admissible representations generated by finitely many vectors. In the next exercise we motivate part of Theorem B.

Exercise 4. (*)² Let $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and $\pi = \mathcal{C}_c^\infty(\mathbb{P}^1(\mathbb{Q}_p))$ be as in the last exercise.

- (a) Show that this is an admissible representation.
- (b) Assume that Theorem A holds for π , where G^{rs} is the subset consisting of $g \in \mathrm{GL}_2(\mathbb{Q}_p)$ with distinct eigenvalues λ_1, λ_2 . Show that

$$t_\pi(g) = \begin{cases} 0 & \text{when } \lambda_1, \lambda_2 \notin \mathbb{Q}_p \text{ (but in a quadratic extension),} \\ \frac{|\lambda_1| + |\lambda_2|}{|\lambda_1 - \lambda_2|} & \text{when } \lambda_1 \neq \lambda_2 \in \mathbb{Q}_p. \end{cases}$$

Here, for general field $\mathcal{C}$, the quantity $\frac{|\lambda_1| + |\lambda_2|}{|\lambda_1 - \lambda_2|} \in \mathbb{Z}[1/p]$ is realized in $\mathcal{C}$ via the unique map $\mathbb{Z}[1/p] \rightarrow \mathcal{C}$.

Remark 2. We see in the last exercise that, when $\mathcal{C} = \mathbb{C}$, the function $t_\pi \cdot |D_G|^{1/2}$ is locally bounded; here $D_G(g) = (\lambda_1 - \lambda_2)^2$. It is not hard, though seemingly cumbersome, to show that t_π is locally integrable. When $\mathrm{char}(\mathcal{C}) = 2$ and p is odd, we have $p = 1$ in $\mathcal{C}$ and consequently $t_\pi \equiv 0$.

¹This sheet makes use of materials reviewed in §2 of the course note. You are only required to hand in 1 or 2 questions **in total** from both sheets.

²Indicating that this problem is more difficult.